UPPER LIMB ANATOMY

EXAM-DETAIL REFERENCE • MUSCLES • BONES • JOINTS • NERVES • CLINICAL CORRELATIONS

Original study text reorganised into a compact, black-text textbook layout. Illustrations are text-free anatomical visual aids; use the written tables for exact attachments, nerve roots and branch origins.

How to use this guide

Learn the regional map first, then memorise attachments and innervation. In practical exams, always orient a bone before naming a feature. In true/false questions, watch for swapped sides, wrong joint surfaces, incorrect root values, and muscles assigned to the wrong compartment.

Important accuracy note. This is a newly compiled study guide, not a verbatim reproduction of Snell or Gray's. Exact page numbers depend on edition and format, so they are not guessed. Common anatomical variations are flagged where relevant. Use your assigned textbook and demonstrator's preferred terminology when an examination syllabus specifies an edition.

Regional map

Region	Must know
Pectoral/anterior axioappendicular	Pectoralis major/minor, subclavius, serratus anterior; clavipectoral fascia; axillary walls.
Scapular/posterior axioappendicular	Trapezius, latissimus dorsi, levator scapulae, rhomboids; dorsal scapular, accessory, thoracodorsal and long thoracic nerves.
Scapulohumeral	Deltoid, teres major, rotator cuff; quadrangular/triangular spaces; shoulder capsule.
Arm	Anterior flexors and posterior extensors; musculocutaneous and radial nerve pathways; profunda brachii.
Forearm	Flexor-pronator and extensor-supinator compartments; median/ulnar/radial branches; interosseous nerves.
Wrist/hand	Carpal tunnel, Guyon canal, snuffbox, palmar arches, intrinsic muscles, extensor expansion.

Terminology trap: the arm is shoulder-to-elbow; the forearm is elbow-to-wrist. 'Upper limb' includes both plus the shoulder girdle and hand.

1. Osteology: bones and small details

First orient each bone. The same landmark can look different in anterior, posterior, medial or lateral views.

Scapula

Feature	Precise detail / trap
Spine	Prominent posterior ridge divides supraspinous and infraspinous fossae; continues laterally as acromion.
Acromion	Articulates with lateral end of clavicle at the acromioclavicular joint.
Coracoid process	Hook-like anterior projection; attachments include short head of biceps and coracobrachialis (common tendon), pectoralis minor insertion, and coracoclavicular/coracoacromial/coracohumeral ligaments.
Glenoid cavity	Shallow, pear-shaped articular surface for humeral head; glenoid labrum deepens it.
Supraglenoid tubercle	Long head of biceps brachii origin.
Infraglenoid tubercle	Long head of triceps brachii origin.
Subscapular fossa	Costal/anterior surface; subscapularis origin.
Supraspinous fossa	Smaller posterior fossa above spine; supraspinatus origin.
Infraspinous fossa	Larger posterior fossa below spine; infraspinatus origin.
Borders	Medial (vertebral) border is long; lateral (axillary) border is thick; superior border is short and contains the suprascapular notch.
Suprascapular notch	Suprascapular nerve usually passes under the superior transverse scapular ligament; artery usually passes over it. 'Army over, Navy under' is a mnemonic, not a substitute for anatomy.
Scapular translucency	Scapula is a flat bone; thin regions can appear translucent on radiographs. Do not confuse normal thinness with a fracture or lytic lesion.

Clavicle

Feature	Precise detail / trap
Shape	S-shaped long bone; medial two-thirds convex anteriorly, lateral one-third concave anteriorly.
Sternal end	Large, roughly quadrangular; articulates with manubrium and first costal cartilage at sternoclavicular joint.
Acromial end	Flattened; articulates with acromion.
Inferior surface	Conoid tubercle and trapezoid line for coracoclavicular ligament; subclavian groove for subclavius; impression for costoclavicular ligament medially.
Ossification	Predominantly intramembranous; one of the first bones to begin ossifying; medial epiphysis is among the last to fuse. Check course-specific timing.
Clinical	Most commonly fractures near the junction of medial two-thirds and lateral one-third; medial fragment may elevate from sternocleidomastoid pull.

Humerus, radius, ulna and hand bones

Tiny surface features often become attachment questions. Learn each feature with the structure attached to it.

Humerus

Landmark	Attachment / relationship
Head	Smooth, approximately hemispherical articular surface; faces medially, superiorly and posteriorly; articulates with glenoid cavity.
Anatomical neck	Groove immediately around articular head; capsule attaches around it, with inferior variation.
Surgical neck	Common fracture region; axillary nerve and posterior circumflex humeral vessels are nearby.
Greater tubercle	Superior facets receive supraspinatus (upper), infraspinatus (middle), teres minor (lower).
Lesser tubercle	Subscapularis insertion.

Landmark	Attachment / relationship
Intertubercular sulcus	Long head of biceps tendon runs in it; pectoralis major inserts on lateral lip, latissimus dorsi on floor, teres major on medial lip.
Deltoid tuberosity	Deltoid insertion.
Radial groove	Radial nerve and profunda brachii artery pass posteriorly; groove separates medial and lateral heads of triceps origins.
Medial epicondyle	Common flexor-pronator origin; ulnar nerve passes posterior to it.
Lateral epicondyle	Common extensor-supinator origin; radial nerve divides near this region.
Capitulum	Lateral distal articular surface; articulates with radial head.
Trochlea	Medial spool-shaped articular surface; articulates with trochlear notch of ulna.
Coronoid/olecranon fossae	Anterior coronoid fossa receives coronoid process in flexion; posterior olecranon fossa receives olecranon in extension.

Radius and ulna

Feature	Exam detail
Radial head	Disc-shaped proximal end. Superior fovea articulates with capitulum; circumference articulates with radial notch of ulna and is embraced by annular ligament.
Radial tuberosity	Biceps brachii inserts; bicipitoradial bursa helps reduce friction.
Radial styloid	Distal projection on lateral/thumb side; normally extends farther distally than ulnar styloid.
Ulnar notch of radius	Medial distal radius articulates with head of ulna at distal radioulnar joint.
Ulnar head	Distal end; does not directly articulate with carpal bones because the articular disc intervenes.
Olecranon	Triceps insertion; forms tip of elbow.
Coronoid process	Brachialis inserts on ulnar tuberosity/coronoid region; contributes to trochlear notch.
Ulnar tuberosity	Brachialis insertion.

Feature	Exam detail
Supinator crest/fossa	Supinator attachment; deep radial nerve branch winds around radial neck through supinator.
Interosseous borders	Attachment for interosseous membrane; transmits forces between radius and ulna.
Pronated position	Radius crosses anteriorly over ulna; during supination the bones are approximately parallel.

Carpals, metacarpals and high-yield hand landmarks

Bone/region	Detail that is easy to miss
Carpal bones (proximal row, lateral → medial)	Scaphoid, lunate, triquetrum, pisiform. Distal row: trapezium, trapezoid, capitate, hamate.
Scaphoid	Most commonly fractured carpal; proximal pole blood supply is vulnerable because much supply enters distally. Snuffbox tenderness suggests fracture.
Lunate	Can dislocate volarly and threaten median nerve in carpal tunnel.
Pisiform	Sesamoid bone in flexor carpi ulnaris tendon; palpable on ulnar side of wrist.
Hook of hamate	Attachment for flexor retinaculum; forms lateral boundary of Guyon canal and medial boundary of carpal tunnel region; vulnerable in some racket/bat sports.
Tubercle of trapezium	Attachment to flexor retinaculum; forms lateral boundary of carpal tunnel.
Capitate	Largest carpal; head articulates proximally with lunate.
Metacarpal I	Thumb carpometacarpal joint is saddle-shaped; permits opposition through combined movements.
MCP joints	Condyloid joints; collateral ligaments are taut in flexion, so abduction/adduction is greatest with MCP joints extended.
IP joints	Hinge joints; flexion/extension only.
Anatomical snuffbox	Tendons: APL and EPB anterior/lateral boundary; EPL posterior/medial boundary. Floor: scaphoid and trapezium. Radial artery crosses the floor.
Carpal tunnel	Median nerve + 9 flexor tendons (4 FDS, 4 FDP, 1 FPL). Flexor carpi radialis is in its own compartment, not inside the tunnel; palmaris longus lies superficial to retinaculum.
Guyon canal	Ulnar nerve and artery pass superficial to flexor retinaculum, between pisiform and hook of hamate; not through carpal tunnel.
Extensor compartments	Six dorsal wrist compartments under extensor retinaculum; compartment 1 APL/EPB, 2 ECRL/ECRB, 3 EPL, 4 ED/EI, 5 EDM, 6 ECU.

2. Pectoral region and anterior axioappendicular muscles

These muscles connect the axial skeleton to the upper limb or move/stabilise the scapula. Do not confuse insertion on humerus with insertion on coracoid process or ribs.

Muscle	Origin	Insertion	Nerve / roots	Main actions / exam notes
Pectoralis major	Medial half clavicle; sternum; upper costal cartilages; external oblique aponeurosis	Lateral lip of intertubercular sulcus	Lateral and medial pectoral nerves (C5–T1 contributions)	Adducts and medially rotates arm; clavicular head flexes; sternocostal head extends arm from flexed position. Tendon twists: inferior sternocostal fibres insert more proximally/deeply.
Pectoralis minor	Ribs 3–5 near costal cartilages	Medial border/superior surface of coracoid process	Medial pectoral nerve (usually C8–T1)	Stabilises scapula by drawing it anteriorly/inferiorly; accessory inspiration with scapula fixed.
Subclavius	Junction of 1st rib and its costal cartilage	Inferior surface of middle clavicle (subclavian groove)	Nerve to subclavius (C5–C6)	Anchors/depresses clavicle; may protect subclavian vessels/brachial plexus in clavicular fracture.
Serratus anterior	External surfaces of ribs 1–8/9	Anterior surface of medial border of scapula, especially inferior angle	Long thoracic nerve (C5–C7)	Protracts scapula; holds it against thoracic wall; upward rotation for overhead elevation. Injury → winged scapula; wall push test.
Trapezius	Superior nuchal line, external occipital protuberance, nuchal ligament, spinous processes C7–T12	Lateral clavicle, acromion, spine of scapula	Spinal accessory nerve (CN XI) motor; C3–C4 proprioception	Upper fibres elevate, middle retract, lower depress; upper + lower rotate scapula upward. Weak shrug / shoulder droop with CN XI lesion.

Muscle	Origin	Insertion	Nerve / roots	Main actions / exam notes
Latissimus dorsi	Spinous processes lower thoracic vertebrae, thoracolumbar fascia, iliac crest, lower ribs; sometimes inferior angle	Floor of intertubercular sulcus	Thoracodorsal nerve (C6–C8)	Extends, adducts, medially rotates humerus; powerful in climbing, swimming, crutch use.
Levator scapulae	Transverse processes C1–C4	Medial border above spine/superior angle	Dorsal scapular nerve (C5) and cervical branches (C3–C4)	Elevates scapula; downward rotation.
Rhomboid minor	Nuchal ligament; spinous processes C7–T1	Root of scapular spine	Dorsal scapular nerve (mainly C4–C5)	Retracts and downwardly rotates scapula; fixes it to thoracic wall.
Rhomboid major	Spinous processes T2–T5	Medial border from spine to inferior angle	Dorsal scapular nerve (mainly C4–C5)	Retracts/downwardly rotates scapula; assists fixation.

Pectoral-region fascial/space traps

Clavipectoral fascia encloses subclavius and pectoralis minor; its thickened inferior part forms the suspensory ligament of the axilla. The cephalic vein, thoracoacromial vessels and lateral pectoral nerve pierce the clavipectoral fascia. Pectoralis minor divides the axillary artery into three parts: first part medial/proximal to it, second posterior to it, third lateral/distal to it.

3. Scapulohumeral muscles and rotator cuff

The rotator cuff is SITS. Its tendons reinforce the glenohumeral capsule superiorly, anteriorly and posteriorly; the inferior capsule is relatively weak.

Muscle	Origin	Insertion	Nerve / roots	Action / detail
Deltoid	Lateral third clavicle; acromion; spine of scapula	Deltoid tuberosity of humerus	Axillary nerve (C5–C6)	Middle fibres abduct after initiation; anterior flex/medially rotate; posterior extend/laterally rotate. Axillary nerve travels through quadrangular space with posterior circumflex humeral artery.
Supraspinatus	Supraspinous fossa	Superior facet of greater tubercle	Suprascapular nerve (C5–C6)	Initiates first ~15° of abduction; stabilises humeral head. Tendon passes under acromion; commonly affected in cuff pathology.
Infraspinatus	Infraspinous fossa	Middle facet of greater tubercle	Suprascapular nerve (C5–C6)	Lateral rotation; stabilises humeral head.
Teres minor	Middle part of lateral border of scapula	Inferior facet of greater tubercle	Axillary nerve (C5–C6)	Lateral rotation; weak adduction; cuff muscle.
Subscapularis	Subscapular fossa (costal surface)	Lesser tubercle of humerus	Upper and lower subscapular nerves (C5–C7)	Medial rotation and adduction; major anterior cuff muscle.
Teres major	Posterior inferior angle/lower lateral border of scapula	Medial lip of intertubercular sulcus	Lower subscapular nerve (C5–C6)	Adducts, extends and medially rotates arm; not a rotator cuff muscle.
Coracobrachialis	Coracoid process (with short head of biceps)	Middle third medial humerus	Musculocutaneous nerve (C5–C7)	Flexes/adducts arm; musculocutaneous nerve pierces it.
Biceps brachii long head	Supraglenoid tubercle and superior labrum	Radial tuberosity and bicipital aponeurosis	Musculocutaneous nerve (C5–C6)	Supinates powerfully when elbow flexed; assists flexion. Tendon passes through shoulder joint cavity but is extrasynovial.
Biceps brachii short head	Coracoid process	Radial tuberosity and bicipital aponeurosis	Musculocutaneous nerve (C5–C6)	Flexion/supination; common origin with coracobrachialis.
Brachialis	Distal half anterior humerus	Coronoid process and ulnar tuberosity	Musculocutaneous (C5–C6); small radial contribution possible	Primary elbow flexor regardless of forearm position; not a forearm muscle.
Triceps brachii long head	Infraglenoid tubercle	Olecranon of ulna and forearm fascia	Radial nerve (C6–C8, chiefly C7–C8)	Extends elbow; long head also extends/adducts arm and helps stabilise shoulder.
Triceps brachii lateral head	Posterior humerus superior to radial groove	Olecranon	Radial nerve	Elbow extension; active against resistance.

Muscle	Origin	Insertion	Nerve / roots	Action / detail
Triceps brachii medial head	Posterior humerus inferior to radial groove	Olecranon	Radial nerve	Elbow extension; active in all extensions.
Anconeus	Lateral epicondyle	Lateral olecranon and proximal posterior ulna	Radial nerve (C7–T1)	Assists extension; stabilises elbow and abducts ulna slightly during pronation.

Shoulder spaces, capsule and movement mechanics

Structure	Boundaries / contents / exam point
Quadrangular space	Superior: teres minor; inferior: teres major; medial: long head triceps; lateral: surgical neck humerus. Contents: axillary nerve + posterior circumflex humeral artery.
Triangular space	Superior: teres minor; inferior: teres major; lateral: long head triceps. Contents: circumflex scapular artery.
Triangular interval	Superior: teres major; medial: long head triceps; lateral: humeral shaft/lateral head triceps. Contents: radial nerve + profunda brachii artery.
Glenohumeral capsule	Loose; attaches around glenoid margin/labrum and humeral anatomical neck with inferior extent. Weak inferiorly; stability relies heavily on cuff, labrum, ligaments and surrounding muscles.
Coracoacromial arch	Coracoid + acromion + coracoacromial ligament form protective arch above humeral head; supraspinatus tendon/subacromial bursa pass below.
Abduction	Supraspinatus initiates; deltoid is major abductor after initiation. Full overhead elevation requires scapular upward rotation by trapezius and serratus anterior.
Rotations	Infraspinatus and teres minor laterally rotate; subscapularis, pectoralis major, latissimus dorsi and teres major medially rotate.
Scapulothoracic articulation	A functional gliding interface, not a true synovial joint; scapula glides over thoracic wall.
AC joint	Plane synovial joint; coracoclavicular ligament (conoid + trapezoid) is a major stabiliser. Separation involves disruption of AC/coracoclavicular structures.
SC joint	Saddle-type synovial joint with articular disc; only true bony articulation between upper limb and axial skeleton. Costoclavicular ligament is an important stabiliser.

4. Arm compartments and key relationships

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Arm compartment questions often test the nerve, artery, and the exact location where a nerve changes compartments.

Muscle	Origin	Insertion	Nerve / roots	Main actions / exam notes
Biceps brachii	Long head: supraglenoid tubercle; short head: coracoid process	Radial tuberosity + bicipital aponeurosis	Musculocutaneous C5–C6	Supination and elbow flexion; tendon in intertubercular groove; aponeurosis protects brachial artery/median nerve superficially.
Brachialis	Distal half anterior humerus	Ulnar tuberosity and coronoid process	Musculocutaneous C5–C6; small radial contribution to lateral part	Most consistent elbow flexor; works in pronation and supination.
Coracobrachialis	Coracoid process	Middle medial humerus	Musculocutaneous C5–C7	Flexes/adducts arm; nerve pierces it, a useful landmark.
Triceps long head	Infraglenoid tubercle	Olecranon	Radial C6–C8	Crosses shoulder and elbow; extends arm and forearm.
Triceps lateral head	Posterior humerus superior to radial groove	Olecranon	Radial C6–C8	Origin superior to radial groove.
Triceps medial head	Posterior humerus inferior to radial groove	Olecranon	Radial C6–C8	Deep head; origin inferior to radial groove.
Anconeus	Lateral epicondyle	Lateral olecranon/proximal ulna	Radial C7–T1	Small extensor assistant; sometimes treated with posterior forearm group.

Brachial artery and nerves

The brachial artery begins at the inferior border of teres major as the continuation of the axillary artery and usually divides into radial and ulnar arteries in the cubital fossa. The median nerve accompanies the artery in the arm and crosses from lateral to medial relative to it. The ulnar nerve leaves the anterior compartment by piercing the medial intermuscular septum and passes posterior to the medial epicondyle. The radial nerve travels with profunda brachii in the radial groove before piercing the lateral intermuscular septum.

5. Anterior forearm: flexor-pronator group

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Most superficial/intermediate flexors are median-innervated. Key exceptions: flexor carpi ulnaris and medial half of flexor digitorum profundus are ulnar-innervated.

Muscle	Origin	Insertion	Nerve / roots	Action / trap
Pronator teres	Medial epicondyle; coronoid process	Lateral mid-radius	Median nerve (C6–C7)	Pronates forearm; weak elbow flexion. Median nerve passes between its heads.
Flexor carpi radialis	Medial epicondyle	Bases of 2nd (often 3rd) metacarpals	Median (C6–C7)	Wrist flexion and abduction (radial deviation). Tendon lies lateral to palmaris longus.
Palmaris longus	Medial epicondyle	Flexor retinaculum and palmar aponeurosis	Median (C7–C8)	Tenses palmar aponeurosis; absent in a substantial minority; not inside carpal tunnel.
Flexor carpi ulnaris	Medial epicondyle; olecranon/posterior ulna	Pisiform, hook of hamate, base of 5th metacarpal via ligaments	Ulnar (C7–T1)	Flexes/adducts wrist. Ulnar nerve passes between its heads.
Flexor digitorum superficialis	Medial epicondyle, coronoid process, radius	Sides of middle phalanges digits 2–5	Median (C7–T1)	Flexes PIP joints; also MCP/wrist. Tendons split (Camper chiasm) for FDP to pass.
Flexor digitorum profundus	Proximal ulna and interosseous membrane	Bases of distal phalanges digits 2–5	Lateral half: AIN (median); medial half: ulnar (C8–T1)	Flexes DIP joints. Classic dual nerve supply.
Flexor pollicis longus	Anterior radius and interosseous membrane	Base of distal phalanx thumb	Anterior interosseous nerve (C8–T1)	Flexes thumb IP joint.
Pronator quadratus	Distal anterior ulna	Distal anterior radius	Anterior interosseous nerve (C8–T1)	Prime pronator, especially with elbow flexed; holds radius/ulna together. Deepest distal anterior muscle.

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Anterior forearm: vessels, nerves and tunnels

Structure	Route / branching / clinical trap
Median nerve	Forms from lateral and medial cords; descends with brachial artery, crosses from lateral to medial in arm; enters forearm between heads of pronator teres; passes between FDS and FDP; enters carpal tunnel deep to flexor retinaculum.
Anterior interosseous nerve (AIN)	Motor branch of median nerve in proximal forearm; supplies FPL, lateral FDP (index/middle), pronator quadratus. AIN syndrome causes weak pinch/OK sign without cutaneous sensory loss.
Ulnar nerve at elbow	Passes posterior to medial epicondyle, then between FCU heads; supplies FCU and medial FDP before entering hand.
Ulnar artery	Usually larger terminal branch than radial in some people; passes deep to FCU and gives common interosseous artery, which divides into anterior/posterior interosseous arteries.
Radial artery	Runs deep to brachioradialis distally, then between brachioradialis and FCR tendons at wrist; passes through anatomical snuffbox to deep palmar arch.
Common interosseous artery	Short branch of ulnar artery; divides into anterior and posterior interosseous arteries, which run on corresponding sides of interosseous membrane.
Flexor retinaculum	Attachments include scaphoid tubercle/trapezium laterally and pisiform/hook of hamate medially; roofs carpal tunnel.
Median nerve compression	Carpal tunnel syndrome affects sensation in lateral 3½ digits and thenar motor function; palmar skin over thenar eminence may be spared because palmar cutaneous branch arises proximal and passes superficial to retinaculum.
Ulnar nerve compression	At cubital tunnel/Guyon canal. In Guyon canal lesion, dorsal ulnar hand sensation may be spared because dorsal cutaneous branch arises proximal to canal.
Superficial flexor tendons	FDS tendons split to allow FDP tendons through; both travel within digital synovial sheaths. FPL tendon travels to thumb distal phalanx.

6. Posterior forearm: extensor-supinator group

The radial nerve supplies the posterior compartment through direct branches and its deep branch/posterior interosseous nerve. Remember that brachioradialis is a flexor despite radial innervation.

Muscle	Origin	Insertion	Nerve / roots	Action / trap
Brachioradialis	Proximal 2/3 lateral supracondylar ridge	Distal radius near styloid	Radial nerve (C5–C7)	Elbow flexion strongest in mid-prone position; radial nerve innervation does not make it an extensor.
Extensor carpi radialis longus	Lateral supracondylar ridge	Base of 2nd metacarpal	Radial nerve (C6–C7)	Extends/abducts wrist.
Extensor carpi radialis brevis	Lateral epicondyle/common extensor tendon	Base of 3rd metacarpal	Deep branch radial nerve (C7–C8)	Wrist extension/abduction; commonly involved in lateral epicondylalgia.
Extensor digitorum	Lateral epicondyle	Extensor expansions digits 2–5	Posterior interosseous nerve (C7–C8)	Extends MCP joints and via expansion IP joints; intrinsic hand muscles coordinate IP extension.
Extensor digiti minimi	Lateral epicondyle	Extensor expansion digit 5	Posterior interosseous nerve (C7–C8)	Extends little finger.
Extensor carpi ulnaris	Lateral epicondyle and posterior ulna	Base of 5th metacarpal	Posterior interosseous nerve (C7–C8)	Extends/adducts wrist; tendon groove on distal ulna.
Supinator	Lateral epicondyle, radial collateral/annular ligaments, supinator crest of ulna	Proximal third radius	Deep branch radial nerve (C7–C8)	Supinates forearm; deep radial nerve passes between its heads.
Abductor pollicis longus	Posterior ulna/radius and interosseous membrane	Base of 1st metacarpal	Posterior interosseous nerve (C7–C8)	Abducts/extends thumb at CMC; first dorsal compartment.
Extensor pollicis brevis	Posterior radius and interosseous membrane	Base of proximal phalanx thumb	Posterior interosseous nerve (C7–C8)	Extends thumb MCP; first dorsal compartment with APL.
Extensor pollicis longus	Posterior ulna and interosseous membrane	Base of distal phalanx thumb	Posterior interosseous nerve (C7–C8)	Extends thumb IP; tendon turns around Lister tubercle; third dorsal compartment.
Extensor indicis	Posterior distal ulna and interosseous membrane	Extensor expansion index finger	Posterior interosseous nerve (C7–C8)	Independent extension of index finger.
Anconeus	Lateral epicondyle	Lateral olecranon/proximal posterior ulna	Radial nerve (C7–T1)	Assists elbow extension; stabilises joint.

Posterior forearm: radial nerve branch map

Branch / landmark	What it does / why it is tested
Radial nerve in arm	From posterior cord (C5–T1); passes through triangular interval with profunda brachii; runs in radial groove; pierces lateral intermuscular septum.
At lateral elbow	Divides into superficial branch (primarily sensory) and deep branch (motor). The exact division level varies around the lateral epicondyle.
Deep branch	Passes around radial neck through supinator; emerges as posterior interosseous nerve. Usually motor to most posterior forearm muscles; ECRL and often ECRB receive innervation before/near entry.
Posterior interosseous nerve	Motor nerve; no major cutaneous territory. Lesion may cause finger/thumb extension weakness with preserved wrist extension (often radial deviation due to ECRL/ECRB spared).
Superficial radial nerve	Descends under brachioradialis; becomes superficial in distal forearm; supplies dorsolateral hand and proximal dorsal parts of lateral digits, not the nail beds of the lateral digits (median territory).
Radial nerve at spiral groove	Humeral shaft fracture may cause wrist drop; triceps often relatively spared depending on lesion level; sensory deficit can involve dorsal first web space.
Posterior interosseous syndrome	Finger extension weakness without sensory loss; differentiate from a high radial nerve lesion, which can impair wrist extension and sensation.
Radial tunnel/arcade of Frohse	Fibrous proximal edge of supinator can compress deep branch; pain may occur without the classic motor deficit of posterior interosseous palsy.

7. Intrinsic hand muscles

Muscle/group	Origin	Insertion	Nerve / roots	Action / exam trap
Abductor pollicis brevis	Flexor retinaculum, scaphoid and trapezium tubercles	Lateral base of proximal phalanx thumb	Recurrent branch median (C8–T1)	Abducts thumb perpendicular to palm; thenar eminence.
Opponens pollicis	Flexor retinaculum and trapezium	Lateral 1st metacarpal	Recurrent branch median	Opposition: flexion + medial rotation of 1st metacarpal at CMC.
Flexor pollicis brevis	Flexor retinaculum/trapezium (superficial head); deep carpal bones (deep head)	Base of proximal phalanx thumb	Superficial head usually median; deep head often deep ulnar	Flexes thumb MCP. Nerve supply can vary; do not oversimplify all heads as median.
Adductor pollicis	Bases of 2nd/3rd metacarpals and capitate (oblique head); shaft 3rd metacarpal (transverse head)	Medial base of proximal phalanx thumb	Deep branch ulnar (C8–T1)	Adducts thumb; Froment sign uses FPL substitution if weak.
Abductor digiti minimi	Pisiform and FCU tendon	Medial base of proximal phalanx digit 5	Deep branch ulnar	Abducts little finger.
Flexor digiti minimi brevis	Hook of hamate and flexor retinaculum	Base proximal phalanx digit 5	Deep branch ulnar	Flexes MCP digit 5.
Opponens digiti minimi	Hook of hamate/flexor retinaculum	Medial border 5th metacarpal	Deep branch ulnar	Draws 5th metacarpal anteriorly and rotates it for cupping palm.
Palmaris brevis	Palmar aponeurosis/flexor retinaculum	Skin of ulnar palm	Superficial branch ulnar	Wrinkles skin and deepens hollow of palm; superficial, not one of deep hypothenar muscles.
Lumbricals 1–2	Lateral sides of FDP tendons	Lateral extensor expansions digits 2–3	Median nerve	Flex MCP while extending IP joints.
Lumbricals 3–4	Adjacent sides of FDP tendons	Lateral extensor expansions digits 4–5	Deep branch ulnar	Same action; bipennate lumbricals 3–4.

Muscle/group	Origin	Insertion	Nerve / roots	Action / exam trap
Dorsal interossei (4)	Adjacent sides of metacarpals	Bases proximal phalanges/extensor expansions digits 2–4	Deep branch ulnar	Abduct digits from axial line through digit 3 (DAB); also flex MCP/extend IP.
Palmar interossei (usually 3)	Palmar surfaces of metacarpals 2, 4, 5	Bases proximal phalanges/extensor expansions	Deep branch ulnar	Adduct digits toward digit 3 (PAD); also flex MCP/extend IP.
Thenar pattern	—	—	Median: APB, OP, usually superficial FPB; ulnar: adductor pollicis, often deep FPB	LOAF is a useful shortcut, not a complete substitute for muscle-by-muscle supply.
Hypothenar pattern	—	—	Ulnar nerve, mainly deep branch except palmaris brevis	Most intrinsic hand muscles are ulnar supplied.

Hand mechanics, extensor expansion and sensory maps

Structure	Fine detail
Extensor expansion	Extensor digitorum tendon spreads into hood over MCP; lumbricals/interossei insert into it. Intrinsic muscles flex MCP and extend IP joints through the hood.
Central axis	Digits abduct/adduct relative to middle finger (digit 3), not the thumb.
Dorsal interossei	Four muscles; digit 3 has one on each side and is not abducted overall by one single muscle.
Palmar interossei	Usually three; no palmar interosseous to digit 3 because it is the reference axis.
Froment sign	Weak adductor pollicis from ulnar neuropathy; patient flexes thumb IP using median-innervated FPL to hold paper.
Ulnar claw	More marked at digits 4–5 with distal ulnar lesion; proximal lesions may produce less clawing due to FDP weakness (ulnar paradox).
Median nerve lesion	Thenar weakness/opposition loss; ape-thumb posture may occur with severe recurrent branch injury; sensory loss in lateral 3½ palmar digits.
Radial sensory area	Dorsal first web space is a useful autonomous test area for radial nerve; do not confuse peripheral nerve territories with dermatomes.
Median sensory area	Index fingertip/palmar distal phalanx is a useful median nerve test area.
Ulnar sensory area	Little fingertip is a useful ulnar nerve test area.
Digital nerves	Proper palmar digital nerves run along sides of digits; avoid casual assumptions that all dorsal distal phalanges are radial sensory territory.

8. Brachial plexus — roots, trunks, divisions, cords and branches

Classic arrangement is C5–T1, but prefixed/postfixed plexuses and other variations exist. Roots are anterior rami, not spinal nerves before the dorsal rami have separated.

Branches by level

Level	Branch	Key distribution / exam note
Roots	Dorsal scapular (C5)	Rhomboids and usually levator scapulae; may pierce middle scalene.
Roots	Long thoracic (C5–C7)	Serratus anterior; descends on superficial surface of muscle on lateral thoracic wall; vulnerable in axillary surgery.
Upper trunk	Suprascapular (C5–C6)	Supraspinatus and infraspinatus; passes through suprascapular notch under ligament; artery usually over ligament.
Upper trunk	Nerve to subclavius (C5–C6)	Subclavius; may give accessory phrenic contribution.
Lateral cord	Lateral pectoral (C5–C7)	Pectoralis major; can communicate with medial pectoral nerve.
Lateral cord	Musculocutaneous (C5–C7)	Anterior arm flexors; pierces coracobrachialis; continues as lateral cutaneous nerve of forearm.
Lateral cord	Lateral root of median nerve	Joins medial root, often anterior to axillary artery.
Medial cord	Medial pectoral (C8–T1)	Pectoralis minor and major; commonly pierces pectoralis minor.
Medial cord	Medial cutaneous nerve of arm (C8–T1)	Skin of medial arm; communicates with intercostobrachial nerve in axilla.
Medial cord	Medial cutaneous nerve of forearm (C8–T1)	Skin medial forearm.

Level	Branch	Key distribution / exam note
Medial cord	Ulnar nerve (C8–T1, chiefly)	FCU, medial FDP, most intrinsic hand muscles; passes behind medial epicondyle.
Medial cord	Medial root of median nerve	Joins lateral root around axillary artery.
Posterior cord	Upper subscapular (C5–C6)	Upper part of subscapularis.
Posterior cord	Thoracodorsal (C6–C8)	Latissimus dorsi; vulnerable in axillary dissection.
Posterior cord	Lower subscapular (C5–C6)	Lower subscapularis and teres major.
Posterior cord	Axillary (C5–C6)	Deltoid and teres minor; through quadrangular space; around surgical neck.
Posterior cord	Radial (C5–T1)	Posterior arm/forearm; triangular interval, radial groove, lateral septum; divides near elbow.
Terminal union	Median nerve (C6–T1 usual)	Most anterior forearm muscles; thenar muscles and lateral two lumbricals. Root values vary by source/individual.

Plexus clinical patterns and branch relationships

Lesion / landmark	Likely findings / trap
Upper trunk (Erb pattern)	C5–C6 traction; weakness of shoulder abduction/lateral rotation and elbow flexion/supination; 'waiter's tip' is a classic simplified pattern.
Lower trunk (Klumpke pattern)	C8–T1 traction, often arm hyperabduction; intrinsic hand weakness/clawing; possible Horner syndrome if sympathetic fibres involved.
Long thoracic nerve	Serratus anterior paralysis → medial winging, impaired upward rotation, difficulty elevating arm above horizontal; wall push accentuates.
Spinal accessory nerve	Trapezius weakness; shoulder droop, weak shrug, impaired upward rotation; often at risk in posterior triangle procedures.
Suprascapular nerve	Weak initiation of abduction and lateral rotation; lesion at notch can affect supra + infra; spinoglenoid notch lesion may mainly affect infraspinatus.
Axillary nerve	Surgical neck fracture/dislocation; deltoid weakness and sensory loss over lateral shoulder ('regimental badge').
Musculocutaneous nerve	Weak elbow flexion/supination; reduced sensation lateral forearm; may be spared partially because brachialis can receive radial contribution.
Median nerve at elbow	Weak pronation, wrist/finger flexion pattern; thenar weakness; sensory changes in lateral hand. AIN lesion is motor-only.
Ulnar nerve at medial epicondyle	Weak finger abduction/adduction, impaired key pinch, sensory changes medial 1½ digits; FCU/medial FDP may be involved.
Radial nerve in groove	Wrist/finger extension weakness; triceps may be largely spared; sensory loss over dorsal first web space.
Axillary artery relations	Cord names are relative to second part of axillary artery. First part medial to pectoralis minor; second behind; third lateral.
Scapular anastomosis	Suprascapular, dorsal scapular and circumflex scapular arteries can provide collateral pathways around scapula. Do not assume all are branches of the same parent artery.

9. Articulations of the upper limb

Joint	Type / surfaces	Ligaments and stabilisers	Movements / exam notes
Sternoclavicular	Saddle-type synovial with articular disc; clavicle with manubrium and 1st costal cartilage	Anterior/posterior SC, interclavicular, costoclavicular ligaments	Elevation/depression, protraction/retraction, axial rotation; only direct bony link of upper limb to axial skeleton.
Acromioclavicular	Plane synovial; acromion with lateral clavicle	AC ligament; coracoclavicular (conoid + trapezoid) stabilises vertical relation	Small gliding/rotation accompanying scapular movement.
Glenohumeral	Ball-and-socket; humeral head with glenoid cavity/labrum	Capsule, superior/middle/inferior GH ligaments, coracohumeral ligament, coracoacromial arch; cuff dynamic stability	Flex/ext, abduction/adduction, medial/lateral rotation, circumduction. Most mobile, relatively unstable.
Elbow humeroulnar	Hinge; trochlea with trochlear notch	Ulnar collateral ligament	Flexion/extension.
Elbow humeroradial	Capitulum with radial head	Radial collateral complex and annular ligament indirectly stabilise radius	Flexion/extension; radial head participates in rotation.
Proximal radioulnar	Pivot; radial head circumference with radial notch of ulna	Annular ligament; quadrate ligament	Pronation/supination.
Distal radioulnar	Pivot; ulnar head with ulnar notch of radius; articular disc	TFCC, palmar/dorsal radioulnar ligaments	Pronation/supination; distal radius moves around ulna.
Radiocarpal (wrist)	Ellipsoid; distal radius + articular disc with scaphoid, lunate, triquetrum	Palmar/dorsal radiocarpal and collateral ligaments	Flex/ext, radial/ulnar deviation, circumduction. Ulna not directly in contact with carpal row.
Intercarpal	Mostly plane synovial	Interosseous/palmar/dorsal ligaments	Gliding contributes to wrist movement.
1st carpometacarpal	Saddle; trapezium with base of 1st metacarpal	Capsule and ligaments; mobility allows opposition	Flex/ext, ab/adduction, circumduction, opposition/reposition.
Other CMC	Plane synovial (2nd–5th)	Strong palmar/dorsal/interosseous ligaments	2nd–3rd relatively fixed; 4th–5th more mobile for cupping.
MCP	Condyloid; metacarpal heads with proximal phalanges	Collateral ligaments, palmar plates, deep transverse metacarpal ligament	Flex/ext, ab/adduction when extended, circumduction.
PIP/DIP and thumb IP	Hinge synovial	Collateral ligaments and palmar plates	Flexion/extension. Thumb has one IP joint, not PIP and DIP.

10. Arteries, veins and lymphatic landmarks

Structure	Route / relation / clinical detail
Axillary artery	Continuation of subclavian at lateral border of rib 1; becomes brachial at inferior border of teres major. Pectoralis minor divides it into three parts.
1st part branches	Superior thoracic artery (classically).
2nd part branches	Thoracoacromial trunk and lateral thoracic artery (classically).
3rd part branches	Subscapular artery, anterior circumflex humeral, posterior circumflex humeral (classically). Branching patterns vary.
Brachial artery	Runs in medial bicipital groove with median nerve; palpable in cubital fossa medial to biceps tendon; divides into radial and ulnar arteries.
Profunda brachii	Travels with radial nerve in radial groove; contributes to elbow anastomosis.
Radial artery	Pulse lateral to FCR tendon at wrist; enters snuffbox; contributes mainly to deep palmar arch.
Ulnar artery	Travels with ulnar nerve distally; contributes mainly to superficial palmar arch; gives common interosseous artery.
Palmar arches	Superficial arch is usually mainly ulnar with radial contribution; deep arch mainly radial with ulnar contribution. Variations common.
Cephalic vein	Lateral superficial vein; ascends deltopectoral groove and pierces clavipectoral fascia to drain into axillary vein.
Basilic vein	Medial superficial vein; pierces deep fascia in arm and joins brachial veins to form axillary vein near inferior border of teres major.
Cubital fossa	Boundaries: brachioradialis lateral, pronator teres medial, line between epicondyles superior. Floor brachialis and supinator; roof fascia/aponeurosis. Contents lateral→medial: biceps tendon, brachial artery, median nerve (TAN). Radial nerve lies lateral.
Lymph	Superficial lymph follows superficial veins; most upper-limb lymph reaches axillary nodes. Lateral hand/forearm often follows cephalic vein to deltopectoral/infraclavicular nodes then apical axillary nodes.

11. Clinical correlations and common traps

Question-style trap	Correct anatomical point
'All rotator cuff muscles insert on greater tubercle.'	False: subscapularis inserts on lesser tubercle; the other three insert on facets of greater tubercle.
'Teres major is part of SITS.'	False: teres major is not a rotator cuff muscle.
'The ulnar nerve passes through the carpal tunnel.'	False: it passes superficial to flexor retinaculum through Guyon canal.
'Flexor carpi radialis is inside the carpal tunnel.'	False: its tendon has a separate compartment in the retinaculum.
'The radial head is at the distal radius.'	False: radial head is proximal; ulnar head is distal.
'The ulna articulates directly with the carpal bones.'	False: articular disc separates ulna from carpus.
'The median nerve supplies every anterior forearm muscle.'	False: FCU and medial FDP are ulnar; brachioradialis is radial and is an elbow flexor.
'All posterior forearm muscles are supplied by posterior interosseous nerve.'	Not strictly: brachioradialis, ECRL, anconeus and often ECRB receive radial/deep radial branches before the PIN proper.
'The axillary nerve is injured in midshaft humeral fracture.'	Classically surgical neck fracture or shoulder dislocation; radial nerve is classically vulnerable in radial groove/midshaft fracture.
'The radial nerve supplies triceps only after it reaches the spiral groove.'	Branches to triceps arise at different levels; exact patterns vary. Relate lesion level to preserved triceps function.
'Long thoracic nerve runs deep to serratus anterior.'	It descends on the superficial/external surface of serratus anterior on the thoracic wall.
'The dorsal scapular nerve is a branch of upper trunk.'	Usually arises directly from C5 root; anatomical variation exists.
'The recurrent median branch supplies all thenar muscles.'	False: adductor pollicis and usually deep head FPB are ulnar supplied; palmaris brevis is superficial ulnar.
'Dorsal interossei adduct fingers.'	False: dorsal abduct (DAB); palmar adduct (PAD), relative to digit 3.
'The thumb has DIP and PIP joints.'	False: thumb has one interphalangeal joint.
'Biceps brachii is the primary elbow flexor in every position.'	False: brachialis is the most consistent/primary elbow flexor; biceps is a powerful supinator.
'Scapulothoracic joint is a synovial joint.'	False: functional articulation, not a true synovial joint.
'The clavicle has a medullary cavity like a typical long bone.'	Its structure/ossification is atypical for a long bone; don't apply generic long-bone assumptions without qualification.
'The humeral head is a perfect hemisphere.'	Textbooks often describe it as approximately hemispherical/roughly one-third of a sphere; avoid absolute geometry claims.
'Median nerve injury always causes sensory loss over the thenar eminence.'	Palmar cutaneous branch arises proximal to tunnel and passes superficial to retinaculum, so classic carpal tunnel may spare thenar skin.

12. Rapid revision: attachment-site map

Bone feature	Structures to recall
Coracoid process	Short head biceps + coracobrachialis origin; pectoralis minor insertion; coracoclavicular, coracoacromial and coracohumeral ligament attachments.
Supraglenoid tubercle	Long head biceps origin.
Infraglenoid tubercle	Long head triceps origin.
Greater tubercle	Supraspinatus, infraspinatus, teres minor.
Lesser tubercle	Subscapularis.
Intertubercular groove lateral lip	Pectoralis major.
Intertubercular groove floor	Latissimus dorsi.
Intertubercular groove medial lip	Teres major.
Radial tuberosity	Biceps brachii insertion.
Ulnar tuberosity/coronoid region	Brachialis insertion.
Olecranon	Triceps insertion; anconeus attachment nearby.
Medial epicondyle	Common flexor-pronator origin; ulnar nerve posterior.
Lateral epicondyle	Common extensor origin; supinator and anconeus origins.
Pisiform	FCU insertion; hypothenar/ulnar ligament attachments.
Hook of hamate	Flexor retinaculum attachment; hypothenar muscle origins.
Scaphoid/trapezium tubercles	Lateral attachment of flexor retinaculum.
Base of 1st metacarpal	APL insertion; opponens pollicis along shaft; thumb CMC joint with trapezium.
Base of 5th metacarpal	ECU insertion; FCU reaches via pisiform/ligaments.

13. Practice statements — answer key included

1. The radial head is disc-shaped and its superior fovea articulates with the capitulum. Answer: True. The circumference also articulates with the radial notch of the ulna and is held by the annular ligament.

2. Subscapularis inserts on the greater tubercle. Answer: False. It inserts on the lesser tubercle; supraspinatus, infraspinatus and teres minor insert on greater-tubercle facets.

3. The ulnar nerve passes posterior to the medial epicondyle before entering the forearm. Answer: True. It then passes between the two heads of flexor carpi ulnaris.

4. The long thoracic nerve supplies trapezius. Answer: False. It supplies serratus anterior; trapezius motor supply is spinal accessory nerve.

5. The ulnar head directly articulates with the lunate. Answer: False. The articular disc intervenes between the ulna and carpus.

6. The median nerve is usually formed by lateral and medial roots from the corresponding cords. Answer: True. The roots commonly unite anterior to the axillary artery.

7. All four rotator cuff muscles laterally rotate the arm. Answer: False. Subscapularis medially rotates; supraspinatus initiates abduction.

8. The posterior interosseous nerve is primarily motor. Answer: True. A PIN lesion can weaken finger/thumb extension without a major cutaneous sensory deficit.

9. The thumb has two interphalangeal joints. Answer: False. It has one IP joint.

10. Flexor carpi ulnaris is supplied by the median nerve. Answer: False. It is supplied by the ulnar nerve.

11. The radial nerve passes through the triangular interval with profunda brachii artery. Answer: True. It then enters the radial groove on the posterior humerus.

12. The scapulothoracic articulation is a true synovial joint. Answer: False. It is a functional gliding interface.

13. The dorsal interossei abduct digits away from the middle finger axis. Answer: True. Palmar interossei adduct toward that axis.

14. The median nerve's palmar cutaneous branch travels through the carpal tunnel. Answer: False. It passes superficial to the flexor retinaculum.

15. The axillary nerve may be injured in surgical-neck fracture of humerus. Answer: True. Look for deltoid weakness and sensory change over lateral shoulder.

16. The long head of biceps originates from the infraglenoid tubercle. Answer: False. Long head biceps: supraglenoid tubercle/superior labrum; long head triceps: infraglenoid tubercle.

17. The cephalic vein pierces the clavipectoral fascia before draining into the axillary vein. Answer: True. It ascends in the deltopectoral groove.

18. The radial artery usually contributes chiefly to the deep palmar arch. Answer: True. The ulnar artery usually contributes chiefly to the superficial arch; variations occur.

19. Brachialis is a more position-independent elbow flexor than biceps brachii. Answer: True. Biceps is especially powerful in supination and also supinates the forearm.

20. The suprascapular nerve usually passes under the superior transverse scapular ligament. Answer: True. The suprascapular artery usually passes over it; variations can occur.

14. Source notes and final verification checklist

Use the named texts to check details against the edition assigned by your course. This guide summarises and cross-checks core regional anatomy but is not a replacement for the complete texts, lab teaching, or a dissection atlas.

Primary references

• Snell’s Clinical Anatomy by Regions, 11th edition (Lawrence E. Wineski, Wolters Kluwer, 2024), Upper Limb chapter. The publisher lists the upper-limb chapter at pp. 93–200 in that edition. Edition pagination differs from older editions. Publisher record.

• Gray’s Anatomy for Students, 2nd/3rd editions, Upper Limb chapter. The Elsevier contents outline covers shoulder, axilla, arm, elbow, forearm, wrist and hand. Publisher record.

• Gray’s Anatomy for Students online textbook excerpt: brachial plexus roots, trunks, divisions, cords, major nerve territories and upper-limb innervation. Online excerpt.

• NCBI Bookshelf / StatPearls, Anatomy, Shoulder and Upper Limb: Radial Nerve; Forearm Muscles; Nerves. These are supplementary summaries, not substitutes for Snell/Gray’s. Radial nerve; Forearm muscles; Nerves overview.

• National Library of Medicine catalog entry for Snell’s Clinical Anatomy by Regions, older 8th edition: bibliographic verification. NLM catalog.

What was checked / what remains edition-sensitive

Core relationships in the tables—muscle attachments, main innervation, nerve pathways, joint surfaces and common clinical lesions—were cross-checked against established upper-limb anatomy descriptions and the sources listed above. Root-value ranges and some muscle nerve branches vary by reference and individual; those are identified as typical or variable where appropriate. Exact textbook page numbers are not supplied unless confirmed for a specific edition.

Illustration note: diagrams are original teaching schematics made for this guide. They simplify spatial anatomy and should be used to orient the topic; written tables and a proper atlas/dissection remain the authority for fine spatial relationships.

Final exam habit: for every nerve, know its roots, origin, course, motor targets, sensory territory and a lesion site. For every bone, orient it first, then identify each surface feature, attachment, articulation and nearby structure.